

Agentic AI-Based Dynamic Tariff Optimization for EV Charging Networks

A three-agent AI framework for demand prediction, dynamic pricing, and continuous performance monitoring

Executive Summary



Demand Prediction Agent

Random Forest model
predicting station utilization
1 hour ahead. $R^2 = 0.9559$,
RMSE = 0.0369 across
425,581 test samples.



Tariff Pricing Agent

Dynamic ₹/kWh pricing
engine with surge (₹22.50)
and discount (₹12.75) rules.
Revenue gain: +11.12%
over fixed ₹15/kWh
baseline.



Monitoring & Learning Agent

144-episode feedback loop
tracking pricing efficiency
(₹16.67/kWh), customer
response (1.13×), and wait
time reduction (0.105).

Data Landscape & Preprocessing

ACN-Data · JPL Site, Caltech

14,848 sessions after cleaning · 27 raw columns
→ 19 engineered features · Coverage: 2018–2019
· Key fields: connectionTime, kWhDelivered, stationID

UrbanEV ST-EVCDP · Shenzhen, China

2,133,833 rows (247 stations × 8,640 timestamps) · 5-minute interval data · 30 days (June 2022) · 9 source files merged into unified base table

Engineered Features

Feature	Formula	Purpose
Utilization Rate	Occupancy / Charger Count	Primary demand signal
Revenue per Session	kWh × ₹15	Baseline revenue proxy
Energy Cost per kWh	₹6 (grid cost assumption)	Profit margin calculation
Queue Length Proxy	Max(0, Occupancy – 80% capacity)	Congestion indicator
Occupancy Density	Occupancy / Station Area	Spatial demand intensity
Demand Zone	<30% discount, 30–80% normal, >80% surge	Pricing label

1,305 ACN rows dropped - null connection timestamps

Sessions >24hrs removed - data entry errors

Negative charge times clipped to 0 - rounding artifacts

EDA: Demand Behavior Insights

Network Is Chronically Underutilized

62%

of station hours below 30% utilization threshold

PRIMARY LEVER: DISCOUNT ACTIVATION, NOT SURGE CONTROL

Demand Peaks Overnight

1am

Peak hour · Lowest at 4pm

FLEET AND RESIDENTIAL OVERNIGHT CHARGING PATTERN

Surge Is Localized, Not Structural

10 stations

drive all surge events · 0 stations average above 80%

SPATIAL REDISTRIBUTION MORE EFFECTIVE THAN BLANKET SURGE PRICING

CBD ≠ Congestion

0.164%

surge rate in CBD vs 1.37% in non-CBD

RESIDENTIAL AREAS DRIVE DEMAND, NOT COMMERCIAL ZONES

Station Level Volatility Justifies Dynamic Pricing

Coefficient of variation ~0.60 across all periods · Peak std=0.194 · Off-peak std=0.152 · 30-day network trend is flat (std=0.009)

Visualizations in appendix: appendix A-F

Demand Prediction Agent

Random Forest · 1-Hour Ahead Forecast · 247 Stations · 8,640 Timestamps

Input Features

Hour, day_of_week,
fast_count, slow_count,
util_lag1, util_lag12,
util_rolling3

Random Forest

100 estimators; Train
1,702,324 rows, Test
425,581 rows

Prediction

Utilization rate 1 hour
ahead

Three Outputs

Predicted utilization,
congestion probability,
expected kWh

Input features flow through a Random Forest model (100 estimators) trained on 1,702,324 rows and tested on 425,581 rows, producing three outputs: predicted utilization rate, congestion probability, and expected charging load (kWh).

RMSE

0.0369

MAE

0.0187

R^2 Score

0.9559

Feature Importance

Feature	Importance
util_lag1	88.6%
hour	3.2%
util_lag12	1.9%
day_of_week	1.4%
station_capacity	3.1%

Agent Output Summary

Congestion flags: **3,693 slots (0.87%)**

Avg expected load: **11.00 kWh per slot**

Max expected load: **108.65 kWh**

Robustness: R^2 improved from 0.67 (no lags) → 0.9559 (with lags + 1hr ahead target)

Tariff Pricing Agent

Rule Based Dynamic Pricing · Demand Elasticity Modeling · Revenue Simulation

Pricing Zones

SURGE ZONE

Utilization $\geq 80\%$ · ₹22.50/kWh · 1.5× multiplier ·

Demand deterrence: -12% sessions

NORMAL ZONE

30% < Utilization < 80% · ₹12.75 → ₹22.50/kWh

· Linear interpolation

DISCOUNT ZONE

Utilization $\leq 30\%$ · ₹12.75/kWh · 0.85× multiplier ·

Demand activation: +35% sessions

Results Metrics

· **+11.12%**
·
·

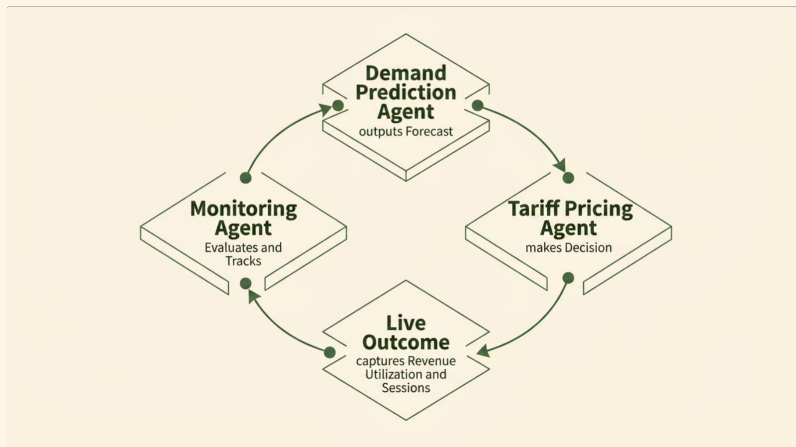
Revenue Gain vs ₹15 Fixed Baseline

- Network utilization: 28.01% → 31.59% (+3.57pp)
- Off-peak session uplift: +35.00%
- Surge utilization: 87.6% → 77.1% (-10.5pp)
- Dynamic Rev/kWh: ₹16.67 vs ₹15.00 baseline

Key finding: Primary revenue driver is off-peak demand activation, not surge premium. 61% of network slots are in discount zone.

Monitoring & Learning Agent

144 Episodes · 1 Episode = 1 Hour of Network-Wide Pricing Decisions



The four agent feedback loop cycles continuously: demand forecasts inform pricing decisions, live outcomes are captured, and the monitoring agent evaluates performance before the cycle repeats.

Pricing Efficiency Score

₹16.67/kWh vs ₹15.00 baseline

+11.1% above benchmark

Customer Response Rate

1.13×

13% more sessions vs fixed
tariff baseline

Avg Wait Time Reduction

0.105 utilization units

Surge zone congestion relief
consistent across all episodes

All 144 episodes above ₹15 baseline · Stable response rate · Consistent surge relief

Business, Operational & Policy Implications

Demand Activation Over Congestion Control

The network is oversupplied - 62% underutilized.

Discount pricing to activate idle capacity generates more value than surge pricing alone.

Idle Time Penalty Pricing

Weekday sessions average 2.69 idle hours post charging. An idle time fee would improve charger turnover without reducing session count.

Spatial Redistribution Policy

10 stations drive all surge events. Directing users to nearby underutilized stations via pricing differentials is more effective than raising prices uniformly.

Location Based Pricing Differentiation

CBD stations underperform non-CBD (0.243 vs 0.293 avg utilization). Tariff design should account for demand topology, not just time of day.

Revenue Sustainability

Dynamic Rev/kWh (₹16.67) exceeds baseline (₹15.00) through volume, not higher unit price. Sustainable model that avoids user backlash from excessive surge fees.

Feedback Loop Deployment

Real deployment requires live session data feeding back to retrain the demand model. Elasticity multipliers should be updated from observed demand response, not assumed.

Limitations & Assumptions

Transparency in modeling choices and boundary conditions

Assumptions

Demand elasticity multipliers (+35% discount, -12% surge) are assumed - not estimated from controlled experiments. A/B testing required for empirical validation.

Grid cost fixed at ₹6/kWh - actual procurement costs vary by time, contract, and location.

Revenue proxy uses utilization as a stand-in for billed kWh - directionally correct but not exact billing figures.

Limitations

ACN (California) and UrbanEV (Shenzhen) datasets are geographically incompatible - used complementarily, not merged into one unified model.

Utilization occasionally exceeds 100% in UrbanEV data - data quality issue flagged, not corrected.

Monitoring agent feedback loop is simulated - real deployment requires live outcome data to retrain agents continuously.

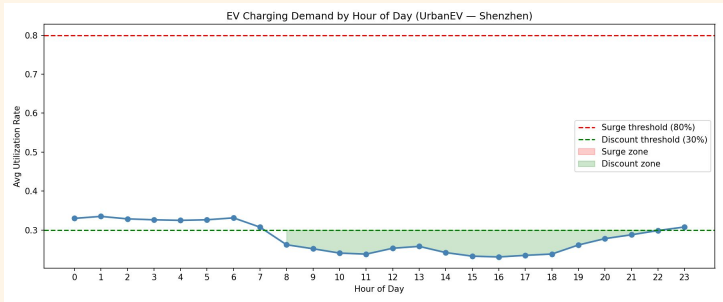
30-day UrbanEV window may not capture seasonal demand variation.



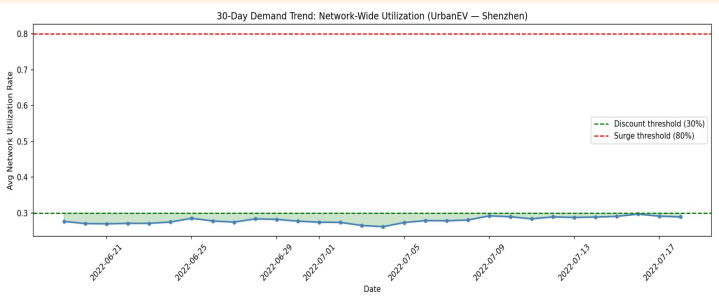
Causal language avoided throughout. All findings are correlational unless explicitly stated otherwise.

Appendix

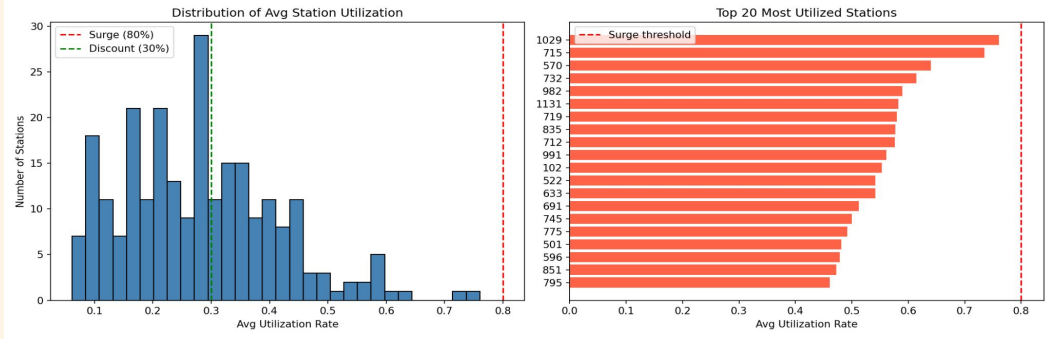
Appendix A: Hourly Demand Curve



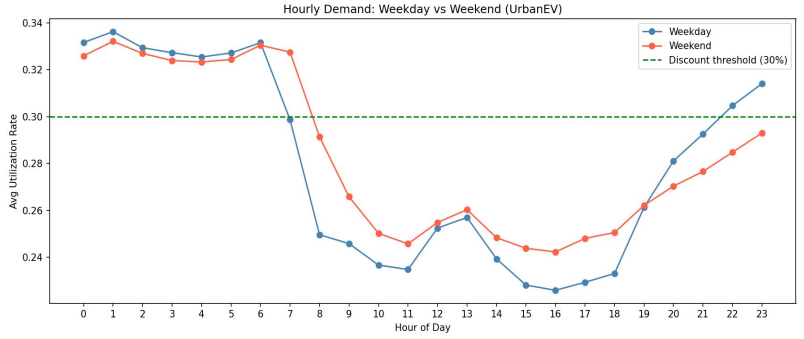
Appendix B: 30-Day Demand Trend



Appendix C: Station Utilisation Distribution

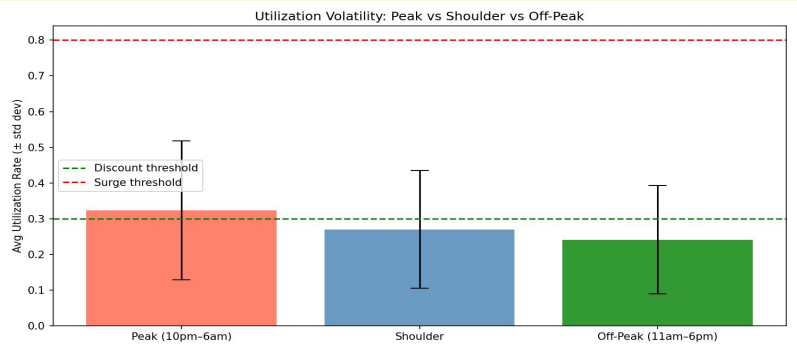


Appendix D: Weekday vs Weekend UrbanEV

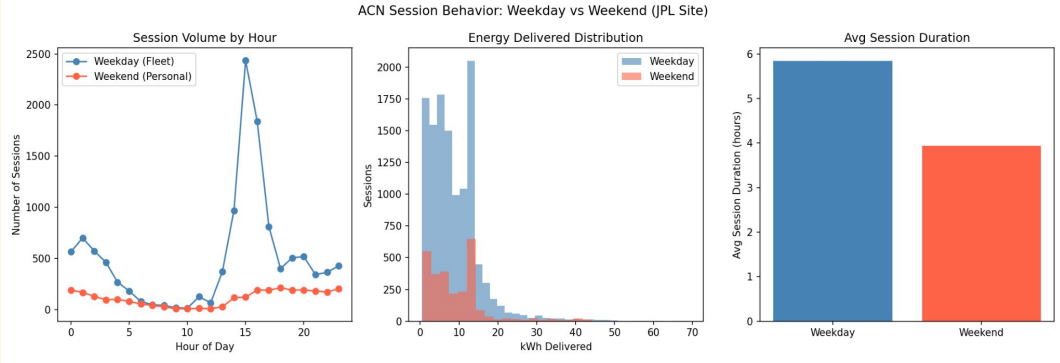


Appendix

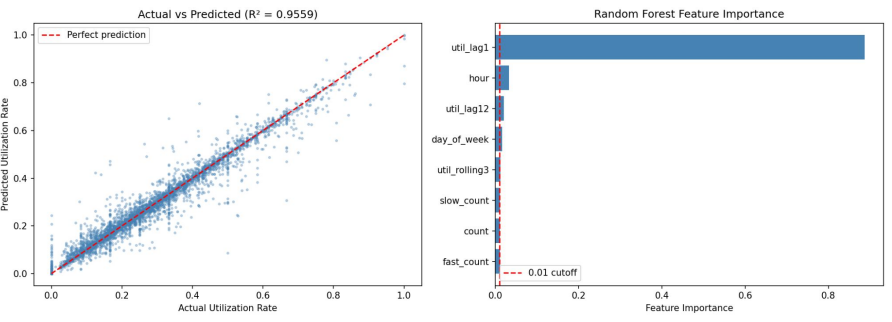
Appendix E: Demand Volatility by Period



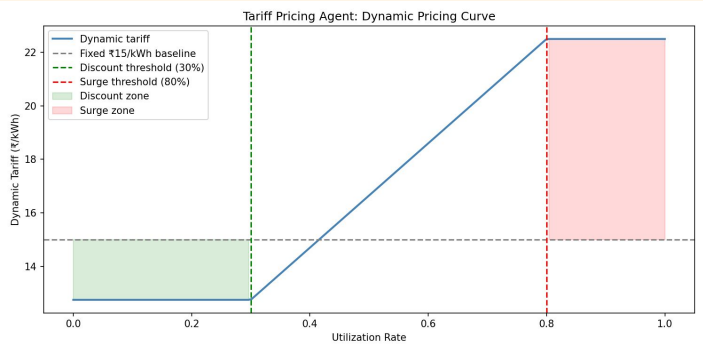
Appendix F: ACN Session Behaviour



Appendix G: Demand Model: Actual vs Predicted

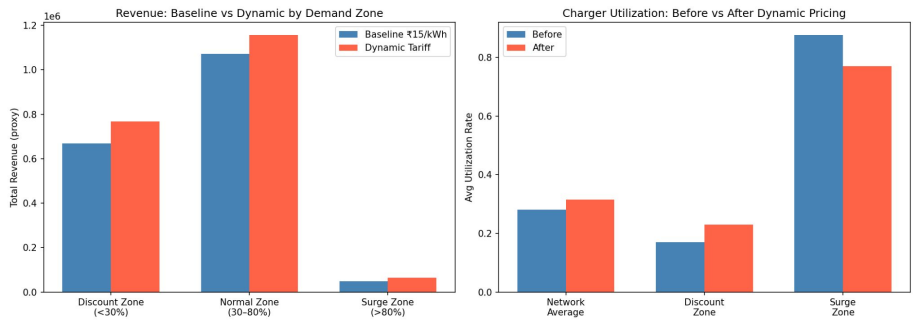


Appendix H: Dynamic Pricing Curve



Appendix

Appendix I: Revenue & Utilization Results



Appendix J: Monitoring Agent Dashboard



Appendix K: Final Metrics Table (final_metrics.csv)

demand_rmse	demand_mae	demand_r2	congestion_flagged_pct	avg_expected_load_kwh	baseline_tariff	surge_tariff	discount_tariff	revenue_gain_pct	network_util_before_pct	network_util_after_pct	util_improvement_pct	offpeak_uplift_pct	surge_util_before	surge_util_after	episodes_tracked	pricing_efficiency_score	customer_response_rate	avg_wait_reduction
0.0369	0.0187	0.9559	0.87	11.0	15.0	22.5	12.75	11.12	28.01	31.59	3.57	35.0	0.876	0.771	144	14.78	1.128	0.1051